

AAPL Historical Stock Analysis — Insight Summary

1. Dataset Overview

This analysis examined Apple Inc. (AAPL) historical stock data from January 2020 to December 2025. The dataset contained 1,508 daily trading records and six main variables: Adjusted Close, Close, High, Low, Open and Volume.

The dataset was inspected for missing values, duplicates, data types and overall structure. The analysis found no missing values and no duplicate records in the dataset.

2. Data Transformation Techniques Applied

Python and Pandas were used to prepare and transform the historical stock data. The analysis included:

- Checking and handling missing values
- Checking for duplicate records
- Sorting the data by date
- Filtering records based on conditions
- Grouping data by year
- Calculating yearly summary statistics
- Extracting year and month information from the date
- Resampling data for weekly and monthly analysis
- Creating calculated columns
- Restructuring the data for analysis and visualization

These transformations converted the raw daily stock data into a format suitable for time-series analysis.

3. Trends Identified

Overall price trend

AAPL's closing price showed a clear overall upward trend between 2020 and 2025, although the movement was not consistently upward. The data showed periods of decline followed by recovery and further growth.

AAPL's closing price increased from \$75.09 on 2 January 2020 to \$271.86 on 31 December 2025, demonstrating substantial long-term price growth despite periods of decline and recovery.

Weekly and monthly performance

Weekly analysis revealed more short-term fluctuations, while monthly analysis provided a clearer view of broader movements within each year.

This shows why looking at both shorter and longer time periods is useful when analyzing historical stock performance.

Moving-average trend

The 7-day and 30-day moving averages reduced daily price fluctuations and made the underlying trend easier to identify.

The 7-day average reacted more quickly to price movements, while the 30-day average provided a smoother view of the broader direction.

Daily returns and volatility

Most daily percentage changes were relatively close to 0%, meaning that large price movements did not occur every trading day.

However, the analysis showed occasional large positive and negative spikes, indicating periods of higher price volatility.

4. Features Created

Feature | Purpose

Daily Price Change | Measures the difference between the opening and closing price

Daily Percentage Change | Measures the daily percentage movement

7-Day Moving Average | Captures short-term price direction

30-Day Moving Average | Captures broader price direction

Monthly Returns | Measures month-to-month percentage performance

30-Day Rolling Volatility | Measures the variability of daily returns over a 30-day window

Daily Price Range | Measures the difference between the day's high and low

These features make the raw stock dataset more useful for monitoring price movement, returns, trends and volatility.

5. Key Findings

Finding 1 — Strong long-term growth

AAPL showed substantial long-term price growth from 2020 to 2025, despite experiencing temporary declines.

Finding 2 — Short-term movements can be significant

The weekly and monthly analysis showed that the overall upward trend contained periods of noticeable short-term fluctuations.

Finding 3 — Moving averages clarify the trend

The 7-day and 30-day moving averages made the underlying price direction easier to interpret by reducing daily noise.

Finding 4 — Most daily returns were relatively small

Daily percentage changes generally remained close to zero, but occasional sharp positive and negative movements showed that AAPL experienced periods of increased volatility.

Finding 5 — Feature engineering improved the analysis

Creating returns, moving averages, price ranges and rolling volatility transformed the raw stock data into indicators that can be used to monitor AAPL's market behaviour more effectively.

6. Recommendations

1. Monitor both short- and long-term price movements.

Use weekly and monthly analysis alongside daily prices to avoid relying on a single time perspective.

2. Use moving averages to monitor trend direction.

The 7-day and 30-day averages can be monitored together to identify changes in short-term and broader price movement.

3. Monitor volatility during periods of sharp price movement.

Daily returns and 30-day rolling volatility can help identify periods where AAPL's price behaviour becomes unusually unstable.

4. Combine price and trading-volume analysis.

Periods of major price movement should be examined alongside trading volume to better understand changes in market activity.

5. Continue updating the engineered features.

Maintaining daily returns, monthly returns, moving averages, price range and volatility measures would support ongoing monitoring of AAPL's performance.

7. Conclusion

The analysis of Apple Inc. (AAPL) historical stock data from 2020 to 2025 shows a strong overall upward price trend accompanied by periods of short-term volatility. Weekly and monthly analysis revealed fluctuations within individual years, while 7-day and 30-day moving averages helped clarify the underlying price direction. Daily percentage changes showed that most movements were relatively small, with occasional sharp gains and losses indicating periods of higher volatility.

Overall, the analysis demonstrates how Pandas data transformation, time-series analysis and feature engineering can turn raw historical stock data into meaningful indicators for understanding price trends, returns and volatility.